WEIGHT 01

Semaglutide

Man-made version of GLP-1, a gut hormone, with 94% of its sequence shared with human GLP-1



CLASS
Peptide (GLP-1 receptor agonist)

STUDIED FOR
Weight, diabetes, heart risk

HUMAN EVIDENCE
Large trials; FDA approved

US STATUS
FDA approved (Wegovy, Ozempic, Rybelsus)

STUDIED FOR

- **Body weight** — HUMAN
- **Blood sugar** — HUMAN
- **Heart events** — HUMAN
- **Liver (MASH)** — HUMAN + ANIMAL
- **Thyroid C-cell tumors** — ANIMAL

Compounded versions are not approved. FDA says the semaglutide shortage ended in Feb 2025. Ozempic and Rybelsus are approved for diabetes; weight use of them is off-label.

HOW IT WORKS IN THE BODY

Semaglutide ▶ GLP-1 receptor ▶ Appetite areas ▶ Food intake

- **Label:** binds and activates the GLP-1 receptor, the target of native GLP-1
- **GLP-1** regulates appetite; its receptor is in brain appetite areas
- **In animals,** it reached and activated neurons that regulate food intake
- **Heart-risk mechanism** in adults has not been established (label)

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- STEP 1: weight -14.9% vs -2.4% at 68 weeks
- SELECT: heart events 6.5% vs 8.0%
- Trials funded by the maker, Novo Nordisk

NOT YET ESTABLISHED

- Thyroid tumor risk in people
- Effects with B12 or glycine

QUICK PRACTICAL INFO

HALF-LIFE
About 1 week (label)

ROUTES STUDIED
Injection under the skin; tablets by mouth

PHARMACOKINETICS
Injection: peak 1 to 3 days; tablets: peak 1 hour (label)

SOLD AS
Prescription (Wegovy, Ozempic, Rybelsus); compounded not approved

MONITORED IN TRIALS
Heart rate, kidney function, blood sugar in diabetes

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved; boxed warning, thyroid C-cell tumors
FDA	Compounded versions not approved
WADA	Not prohibited

REFERENCES

1. FDA, Wegovy prescribing information, Jun 2026
2. N Engl J Med, STEP 1 (Wilding et al.), Mar 2021
3. FDA, Ozempic prescribing information, Jun 2026

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